WEEK 3 · LESSON 13 OF 24 · HANDOUT

Vectors: How AI Sees Data

Study time: **about 45 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=fNk_zzaMoSs — open the lesson online to watch inside the course.

🎓 **Go deeper:** freeCodeCamp: Linear Algebra for Machine Learning — full course — https://www.youtube.com/watch?v=rSjt1E9WHaQ

Your goal in this lesson: Explain what a vector is, add and scale vectors, compute length and the dot product, and use cosine similarity to measure how alike two things are.

Everything becomes a list of numbers

AI cannot read a house, a student or a word directly. It turns each one into a **vector** — an ordered list of numbers:

- A house: `[size_m2, bedrooms, age_years]` → `[120, 3, 10]`
- A student: `[maths, english, science]` → `[78, 64, 85]`
- A word in a language model: hundreds of numbers called an **embedding**.

Each number is one **feature** (one dimension). A vector with 3 numbers lives in 3-dimensional space; an embedding may live in 1,000 dimensions. The maths is the same.

Vectors in NumPy

```
import numpy as np

aline = np.array([78, 64, 85])
eric = np.array([70, 80, 60])

print(aline + eric)        # add element by element
print(aline - eric)        # differences per subject
print(aline * 1.1)         # scale: everyone gets 10% more
print((aline + eric) / 2)  # the average student
```

Geometrically, a 2-D vector is an **arrow** from the origin. Adding vectors puts arrows end to end; scaling makes an arrow longer or shorter.

Length (norm)

The **length** of a vector uses Pythagoras' theorem — square each number, add, take the square root:

```
v = np.array([3, 4])
print(np.sqrt((v ** 2).sum()))   # 5.0
print(np.linalg.norm(v))         # the same, built in
```

The **distance** between two vectors is the length of their difference:

```
print(np.linalg.norm(aline - eric))   # how different the two students are
```

k-nearest neighbours (AI Foundations, Lesson 9) uses exactly this distance to find "similar" examples.

The dot product

Multiply matching elements and add them up:

```
weights = np.array([0.5, 0.2, 0.3])   # importance of each subject
print(np.dot(aline, weights))          # 78×0.5 + 64×0.2 + 85×0.3 = 77.3
print(aline @ weights)                 # @ is the same thing
```

The dot product is the most important operation in AI. **A neuron is a dot product** of inputs and weights, plus a bias. A whole neural network is millions of dot products.

Cosine similarity: same direction?

Two vectors can point in the **same direction** even if one is longer. Cosine similarity measures the angle between them:

> *cosine similarity = (a · b) ÷ (|a| × |b|)*

It is **1** for the same direction, **0** for unrelated (90°), and **−1** for opposite.

```
def cosine(a, b):
    return np.dot(a, b) / (np.linalg.norm(a) * np.linalg.norm(b))

# features: [likes_football, likes_music, likes_maths]
u1 = np.array([5, 1, 4])
u2 = np.array([4, 0, 5])
u3 = np.array([0, 5, 1])
print(round(cosine(u1, u2), 2))   # high -> similar people
print(round(cosine(u1, u3), 2))   # low -> different interests
```

Recommendation systems and search engines use this to find similar users, products and documents. Language models compare word and sentence embeddings the same way.

Key terms

- **Vector** — an ordered list of numbers; one number per feature.
- **Dimension** — the number of values in a vector.
- **Norm (length)** — the size of a vector: square root of the sum of squares.
- **Dot product** — multiply matching elements and add: `a @ b`.
- **Cosine similarity** — how closely two vectors point the same way (−1 to 1).
- **Embedding** — a learned vector that represents something like a word.

Remember this: AI turns everything into vectors. Distance says how different two things are, the dot product powers every neuron, and cosine similarity says how alike two things are.

Name: _____ Date: _____

Activity

Task 1

Make vectors for three students' marks in 4 subjects. Which two students are closest (smallest distance)?

Task 2

Choose weights for the subjects that add to 1 and compute each student's weighted score with `@`.

Task 3

Describe yourself and two friends with 4 interest scores (0–5). Use cosine similarity to find who is most like you.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. What is a vector in AI?

   A. A type of virus
   B. A chart
   C. An ordered list of numbers that describes something, one number per feature
   D. An arrow drawn on paper only

2. What is the length of the vector [3, 4]?

   A. 5
   B. 7
   C. 12
   D. 1

3. What is the dot product of [1, 2] and [3, 4]?

A. 10
B. 7
C. 11
D. [3, 8]

4. Which AI building block is a dot product of inputs and weights (plus a bias)?

A. A histogram
B. A neuron
C. A CSV file
D. A for loop

5. Cosine similarity of two vectors pointing in exactly the same direction is…

A. 1
B. 0
C. 100
D. -1

6. What does the distance between two student vectors tell you?

A. How different their scores are overall
B. Their names
C. Nothing useful
D. Their ages

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